higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL CONSTRUCTION N5

2 DECEMBER 2019

This marking guideline consists of 10 pages.

Copyright reserved

QUESTION 1

1.1 AREAS

Area 1: $15 \times 90 = 1\ 350$

Area 2: $10 \times 20 = 200$

Area 3: $10 \times 20 = 200$

Area 4: $60 \times 8 = 480$

Total area $= 2\ 230$✓ (1)

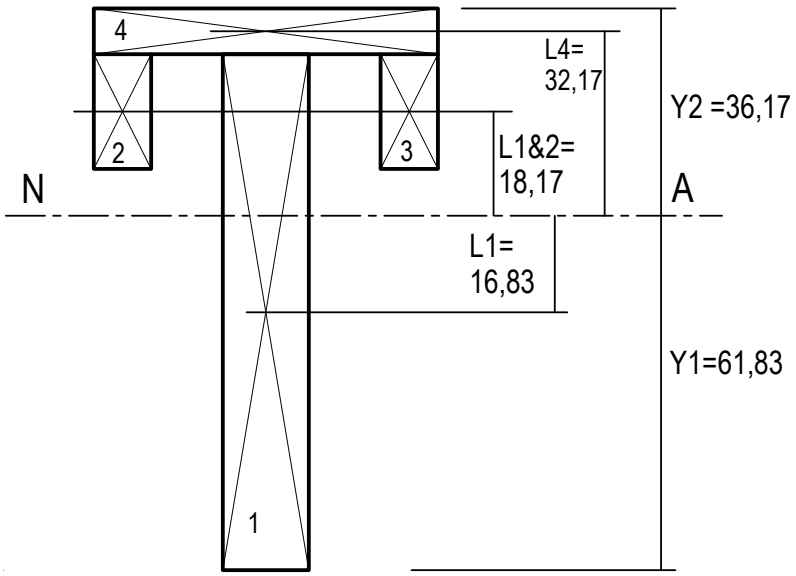
NEUTRAL AXIS FROM A-B

$2\ 230 \text{ x } y_1 = (1\ 350 \times 45) + 2(200 \times 80) + (480 \times 94)$ ✓✓

$Y_1 = \frac{60\ 750 + 32\ 000 + 45\ 120}{2\ 230}$ ✓

$Y_1 = \frac{137\ 870}{2\ 230}$

$Y_1 = 61{,}82 \text{ mm}$ ✓ (4)



1.2 SECOND MOMENT OF AREA

$$I_{xx\,Total} = \left(\frac{bd^3}{12} + AL^2\right) + 2\left(\frac{bd^3}{12} + AL^2\right) + \left(\frac{bd^3}{36} + AL^2\right)$$

$$I_{xx\,1} = \left(\frac{15\ x\ 90^3}{12} + 15 \times 90 \times 16,83^2\right)$$
$$= (911\,250 + 382\,386,02) = 1\,293\,636,02\ mm^4\ ✓✓$$

$$I_{xx\,2\&3} = 2\left(\frac{10\ x\ 20^3}{12} + 10 \times 20 \times 18,17^2\right)$$
$$= 2(6\,666,67 + 66\,029,78) = 145\,392,89\ mm^4\ ✓✓$$

$$I_{xx\,4} = \left(\frac{60\ x\ 8^3}{12} + 60 \times 8 \times 32,17^2\right)$$
$$= (2\,560 + 496\,756,27) = 499\,316,27\ mm^4\ ✓✓$$

I_{xx} TOTAL $= 1\,938\,345,18\ mm^4$ ✓ (7)

1.3 THE SECTION MODULUS (Z) ABOUT THE X-X AXIS

$$Z = \frac{I}{Y}$$

$$Z = \frac{1\ 938\ 345,18\ mm4}{61,83\ mm}\ ✓$$

$Z = 31\,349,59\ mm^3$ ✓ $31,349 \times 10^{-6}\ m^3$ (2)

1.4 BENDING MOMENT (Bending stress = 145 MPa)

$$\frac{m}{I} = \frac{f}{Y}$$

$$M = \frac{1\ 938\ 345,18\ x\ 145}{61,83\ mm}\ ✓$$

$M = 4\,545\,690,62\ Nmm$

(2)

$M = 4,547\ kNm$ ✓ **[16]**

QUESTION 2

2.1 Take moments about RL:

$(RR \times 9) = (9 \times 3 \times 9{,}5) + (28 \times 9) + (7 \times 8 \times 4) + (52 \times 4) + (14 \times 1)$ ✓

$RR = \frac{(256{,}5 + 252 + 224 + 208 + 14)}{9}$ $(\frac{954{,}5}{9})$

$RR = 106{,}06$ kN ✓

Take moments about RR:

$(RL \times 9) + (9 \times 2 \times 1) = (7 \times 8 \times 5) + (14 \times 8) + (52 \times 5) + (9 \times 1 \times 0{,}5)$ ✓

$RL = \frac{(280 + 112 + 260 + 4{,}5) - (18)}{9}$ $(\frac{656{,}5 - 18}{9})$

$RL = 70{,}94$ kN ✓ (4)

2.2

SF A: 70,94 kN	(Max)	70,94 kN ✓
SF A-B: 70,94– (7 x 1)		63,94 kN
SF B: 70,94– (7 x 1) - 14		49,94 kN
SF B-C: 70,94– (7 x 4) - 14		28,94 kN
SF C: 70,94 – (7 x 4) – 14 - 52		-23,06 kN
SF C-D: 70,94 – (7 x 8) – 14 - 52		- 51,06 kN
SF E: 70,94 – (7 x 8) – 14 – 52 – (9 x 1)		- 60,06 kN ✓
SF E: (70,94 + 106,06) – (7 x 8) – 14 - 52– (9 x 1)		46 kN
SF E: (70,94 + 106,06) – (7 x 8) – 14 - 52– (9 x 1) - 28		18 kN
SF F: (70,94 + 106,06) – (7 x 8) – 14 - 52– (9 x 3) - 28		0 (2)

Shear force diagram (See p 6) (2)

2.3 Bending moment diagram (See p 6) (2)

BENDING MOMENT VALUES

BM B: $(70{,}94 \times 1) - (7 \times 1 \times 0{,}5)$ 67,44 kNm✓

BM C: $(70{,}94 \times 4) - (7 \times 4 \times 2) - (14 \times 3)$ (Max) 185,76 kNm✓

BM D: $(70{,}94 \times 8) - (7 \times 8 \times 4) - (14 \times 7) - (52 \times 4)$ 37,52 kNm✓

OR

BM D: $(106{,}06 \times 1) - (9 \times 3 \times 1{,}5) - (28 \times 1)$ 37,52 kNm

BM E: $-(9 \times 2 \times 1)$ -18 kNm✓ (4)

2.4 $Ze = \frac{B\ M\ max}{Stress} = \frac{185{,}76 \times 10^6}{160} = 1\ 161\ 000{,}00\ mm^3$ ✓✓ (2)

$Ze = 1\ 161 \times 10^{-6}\ m^3$ ✓ (1)

2.5 Select: 406 × 178 × 67,2 kg/m I-section parallel flange ($Ze = 1\ 189 \times 10^{-6}$) ✓ (1)

2.6 SHEAR STRESS

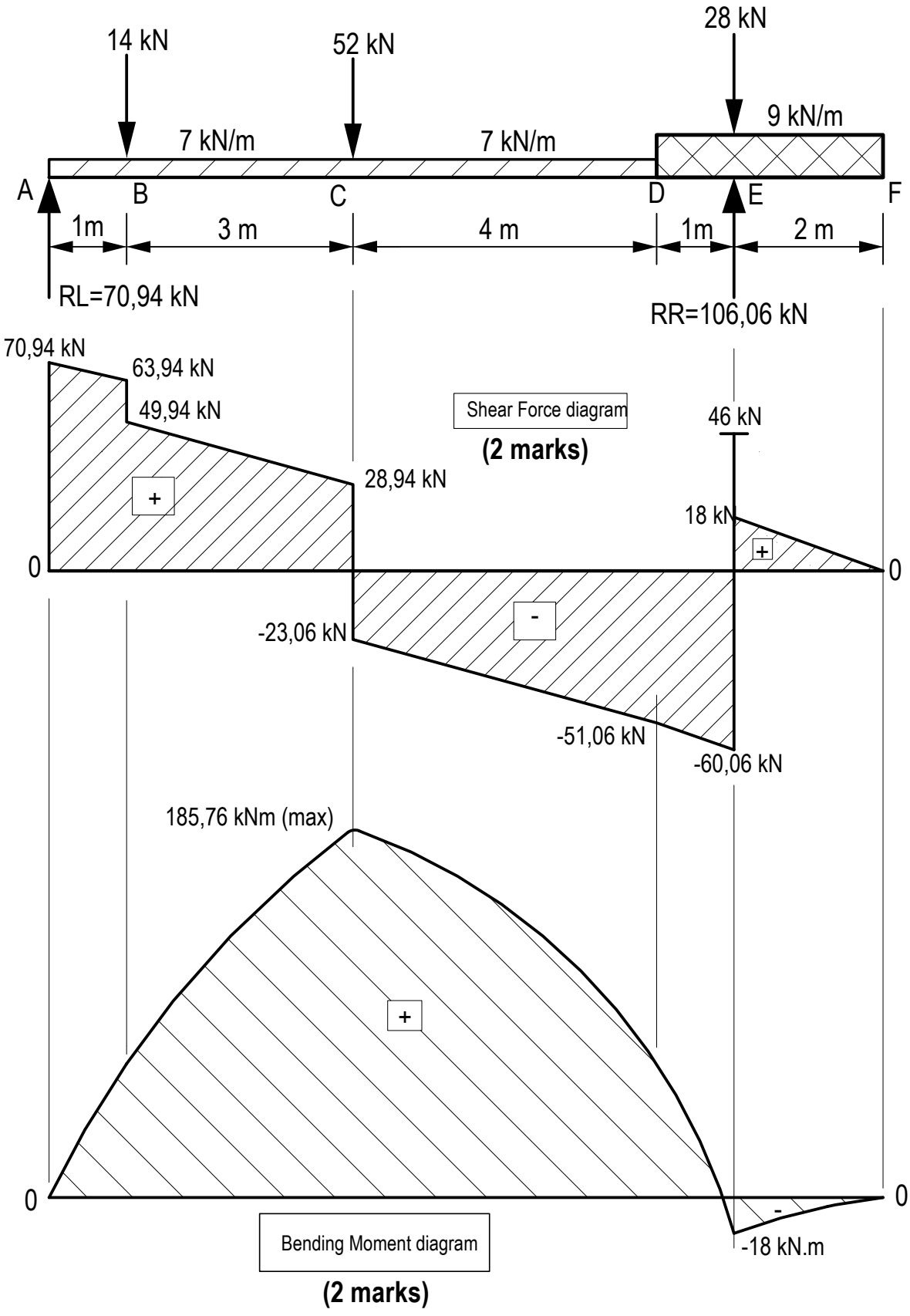
Shear stress $= \frac{S\ F\ max}{D \times t} = \frac{70{,}94 \times 10^3}{409{,}4 \times 8{,}8} = 19{,}69$ MPa ✓✓ (2)

$Ze = 19{,}69 < 100$ MPa✓ (1)

[21]

Space Diagram
Scale: 140 mm = 9 m

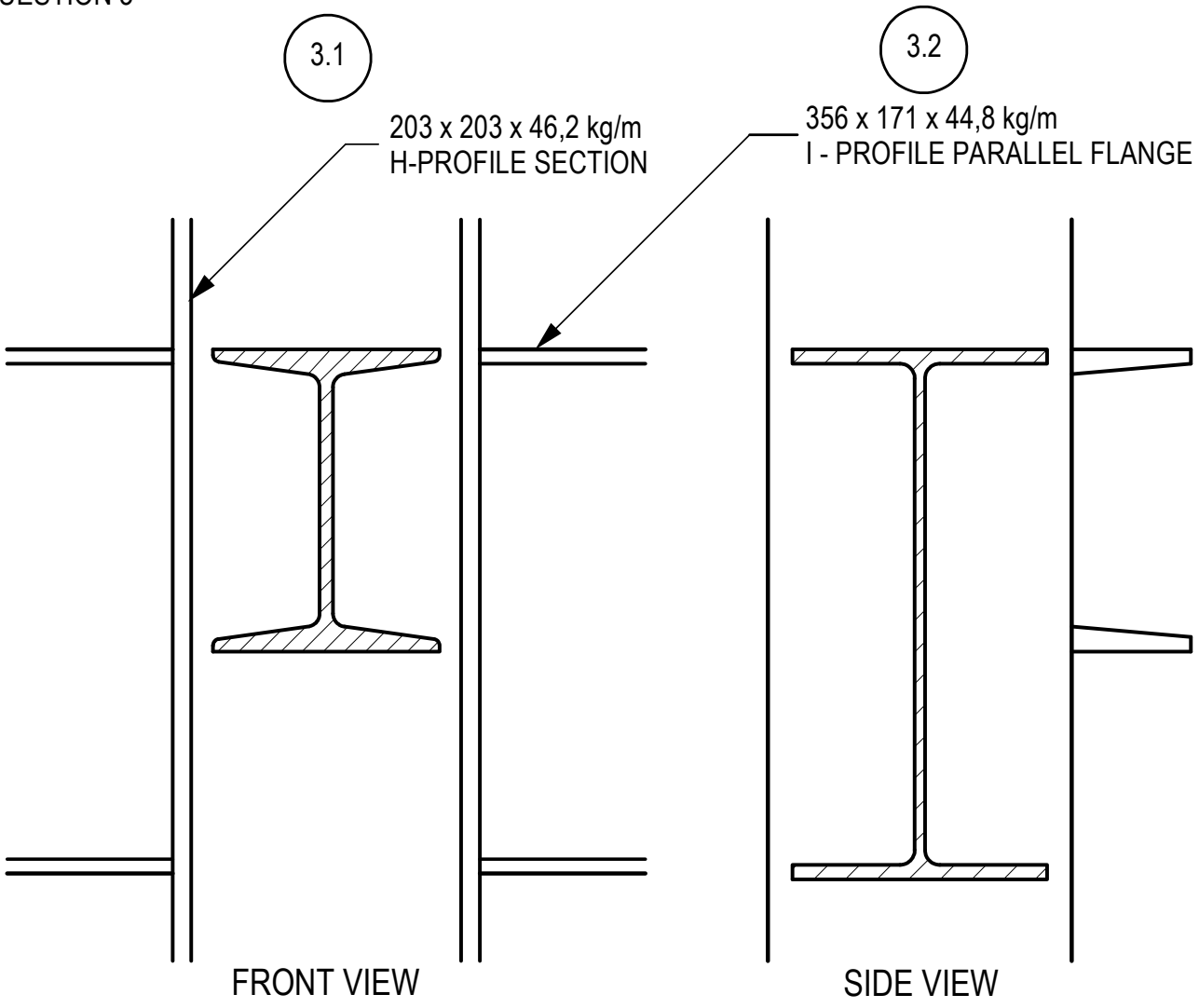
2.2

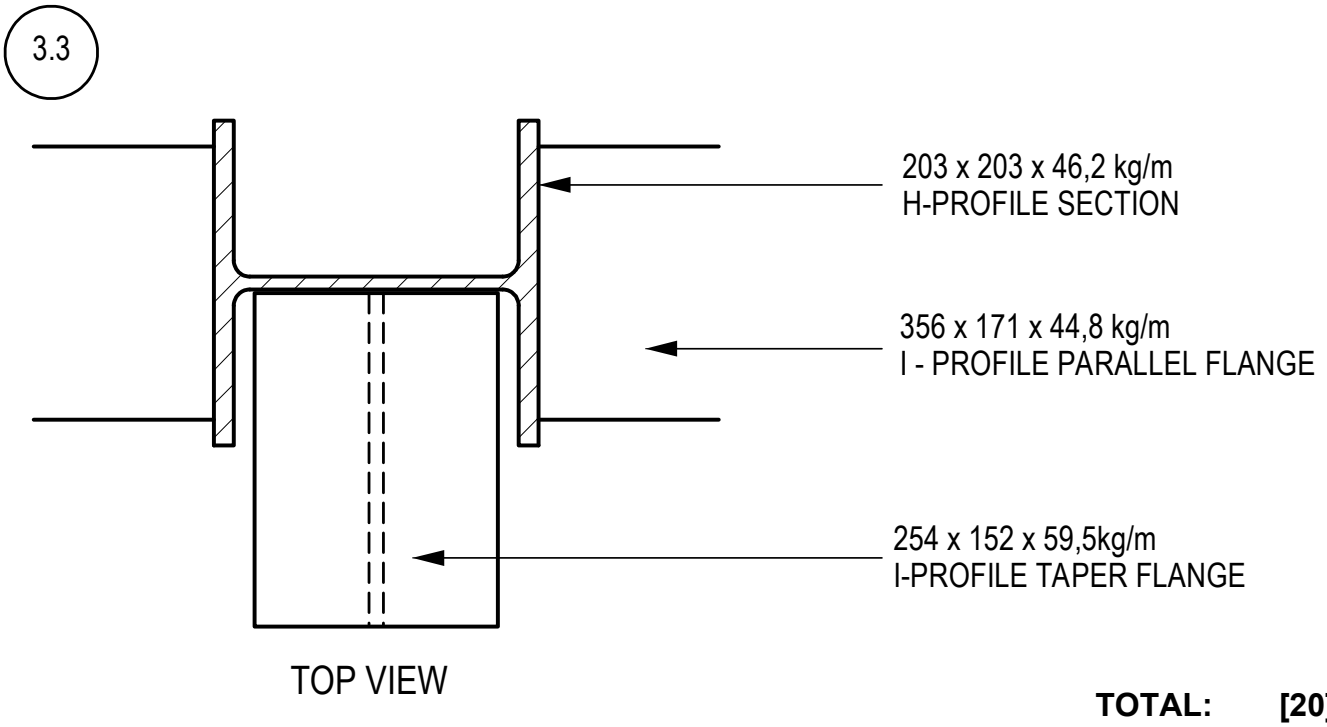


2.3

QUESTION 3

QUESTION 3

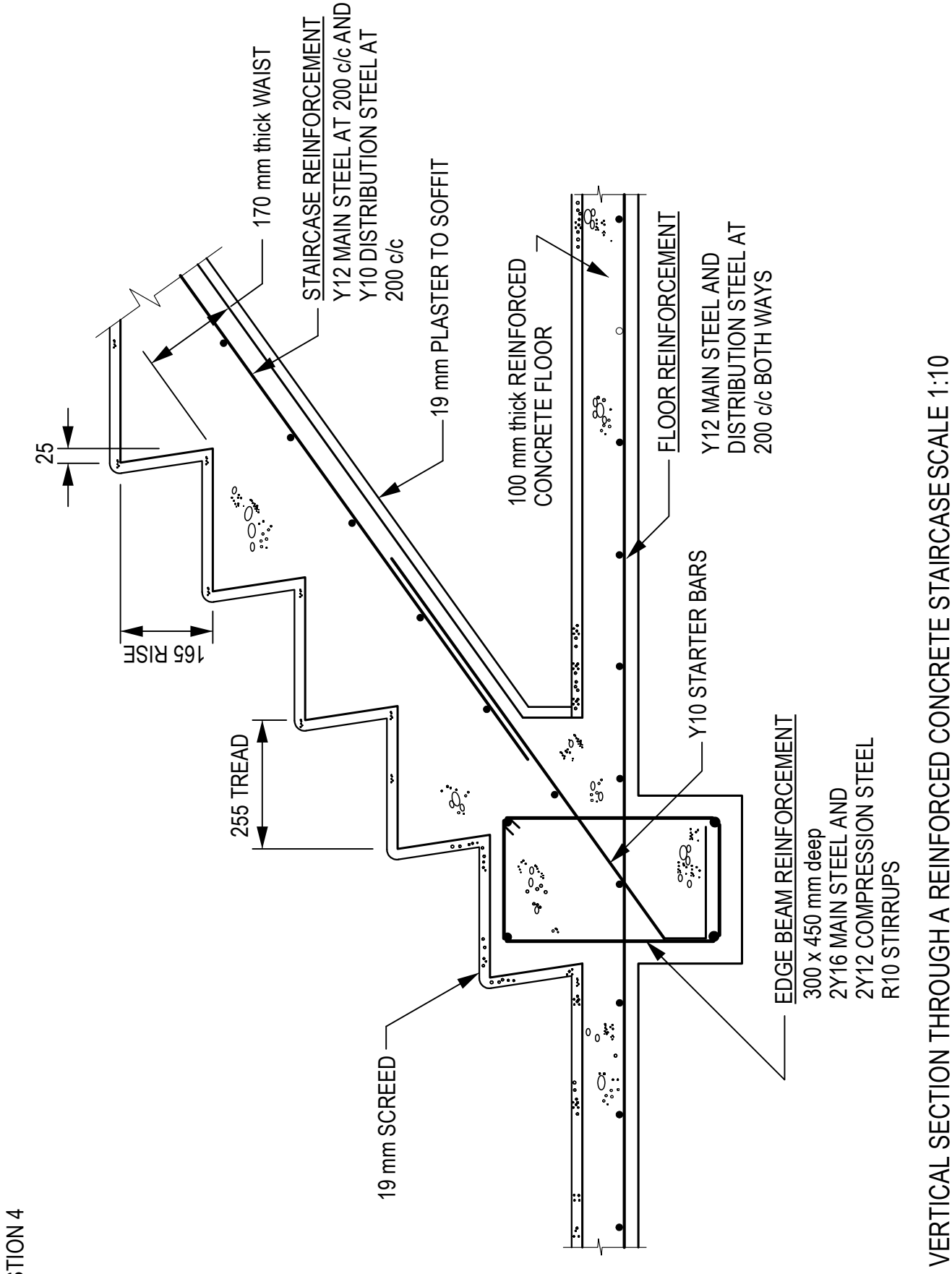




TOTAL: [20]

Copyright reserved

QUESTION 4



VERTICAL SECTION THROUGH A REINFORCED CONCRETE STAIRCASE SCALE 1:10

QUESTION 4

TOTAL: [20]

Copyright reserved

QUESTION 5

1	2	3	4	5	6	7	8	9	10	11	12
Member	Reinforcement					Bending dimensions for shape codes (SANS 282)					
Mark; size & No. off	Mark	Type and size	No. in each	Total No.	Length (mm)	Shape code	A	B	C	D	E or R
Rectangular Beam **2 off**	1	Y25	2	4	6 450	35	6 750				
	2	R20	2	4	6 450	35	6 750				
	3	R20	2	4	6 450	35	6 750				
	4	R16	2	4	6 450	35	6 750				
	5	Y12	2	4	6 450	20	6 450				
	6	Y12	24	48	2 100	60	550	400			
	7	R10	24	48	1 335	72	550	400/3 135			

MARK ALLOCATION

	7 × ½	7 × ½	7 × ½	7 × 1	7 × ½	7 × 1	2 × 1
	3½	3½	3½	7	3½	7	2

Total = 30/2 =15 marks

[15]

QUESTION 6

6.1 E
6.2 F
6.3 A
6.4 C

(4 × 2) **[8]**

TOTAL: 100

Copyright reserved

MARK SHEET FOR QUESTIONS 3 AND 4 (DRAWING PAPER)

QUESTION 3	
MARK ALLOCATION	
Correctness:	
Column	3
Main beam	3
Secondary beam	3
Total	9
9 x 3 Views = 27/3 = 9 marks	

Orthographic projection corr.	2
Accuracy	3
Hatching	2
Lables	2
Titles	2
Total	11

SUMMARY
9 + 11
Total = 20 marks

QUESTION 4		
MARK ALLOCATION		
Correctness:		14
Steps corr. interpretation	2	
Waist	1	
Edge beam corr.	2	
Floor	1	
Screed & Plaster	2	
Beam Reinforcement	2	
Stair Reinforcement	2	
Floor Reinforcement	2	
Accuracy		3
Dimensions		1
Labels		2
Total		20

Copyright reserved